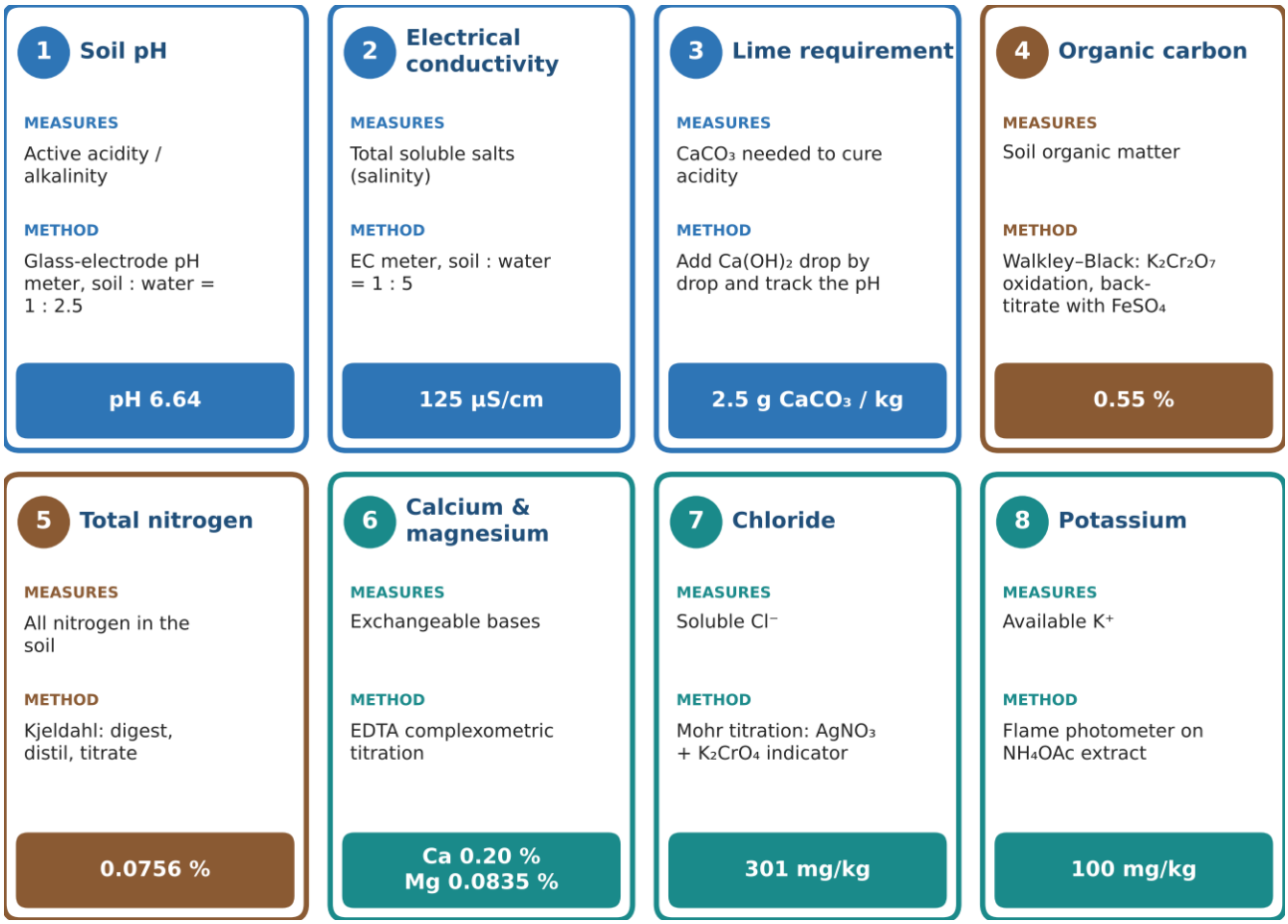
# Soil and Environmental Chemistry

Laboratory Report – Illustrated Study Edition




*The eight experiments at a glance: what each one measures, how, and what we found.*

**How to read this:** This edition keeps the text, data, calculations and results of the original report and adds **apparatus pictures**, **procedure flow-charts** and **concept diagrams** beside each experiment so the work can be understood visually. All drawings are schematic illustrations (not photographs). The Discussion sections are shortened to their key points.

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# Experiment 1: Determination of Soil pH by Using pH Meter

## Objective

To determine the active acidity or alkalinity (pH) of a given soil sample in order to assess its suitability for specific crops and to determine the necessary soil amendments.

## Principle

pH is technically defined as the negative base-10 logarithm of the hydrogen ion concentration expressed in gram equivalents per litre. The pH of soil water is a crucial parameter that indicates the soil's acidity or alkalinity. It is measured on a scale of 0 to 14, where pH 7 is neutral, values below 7 are acidic and values above 7 are alkaline. Monitoring soil pH is essential for assessing its suitability for various biological and agronomic processes.

$$pH = -\log_{10}[H^+]$$

$$H_2O \rightleftharpoons H^+ + OH^- \; ; \; K = [H^+][OH^-] = 10^{-14}$$

A neutral solution has a hydrogen ion concentration of $10^{-7}$, i.e. pH = 7.

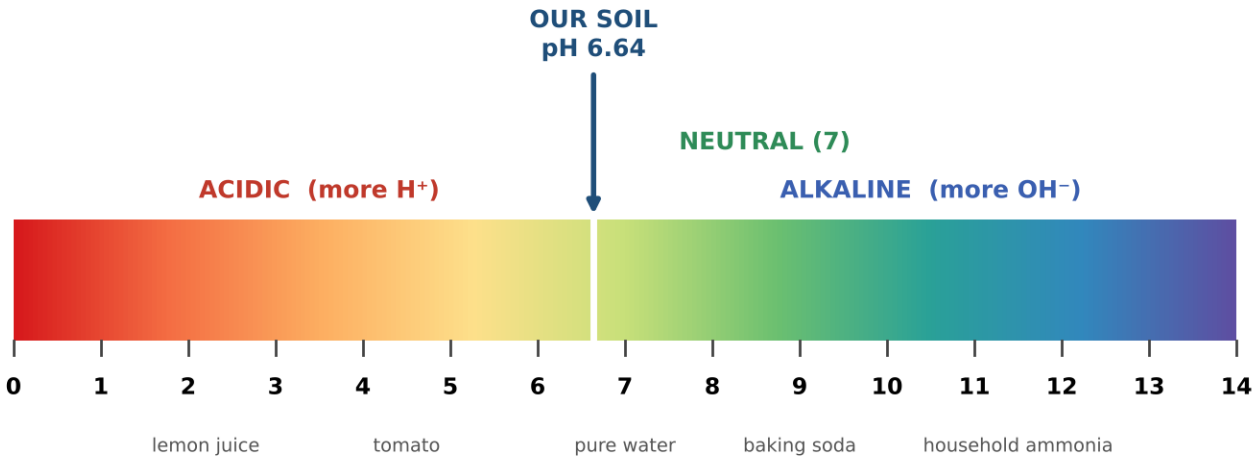



*Fig. 1.1: The pH scale. Each unit is a ten-fold change in $H^+$; our soil (6.64) is almost neutral.*

## Apparatus

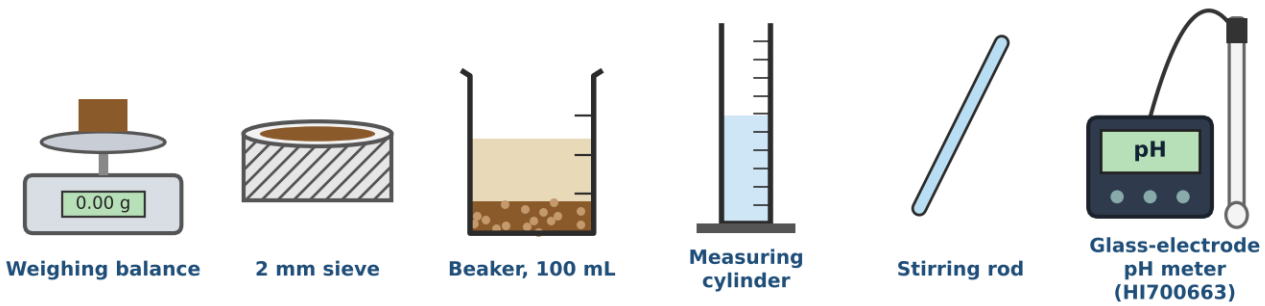



*Fig. 1.2: Apparatus used (schematic).*

## Procedure

1. The soil sample was collected from the laboratory.
2. The sample was ground and passed through a 2 mm sieve.
3. 10 g of soil was weighed into a 100 mL beaker using a weighing balance, and 25 mL of distilled water was added.
4. The suspension was stirred several times over the next 30 minutes.
5. The pH meter was switched on and allowed to stabilise.

**6.** The electrode was placed into the soil suspension, and the reading was recorded once the display had stabilised.

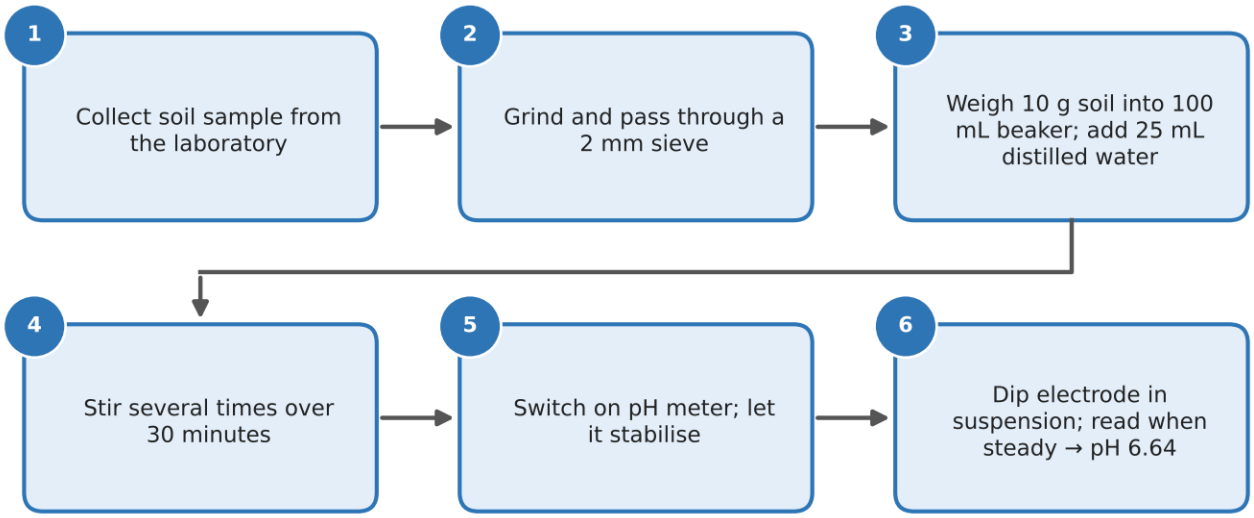



*Fig. 1.3: Procedure at a glance.*

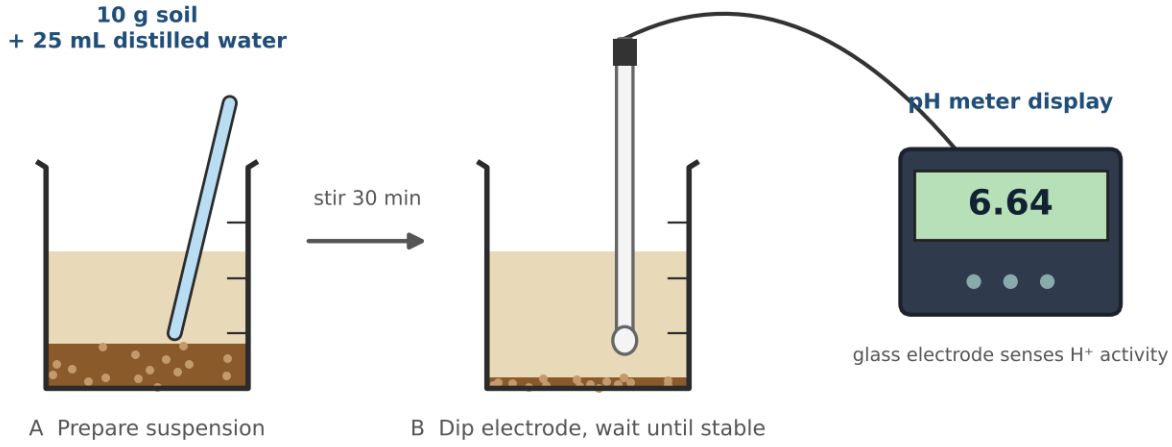



*Fig. 1.4: Setting up the measurement.*

## Observation

The stabilised reading obtained from the pH meter for the given soil sample was: **pH = 6.64**

## Result

**Result:** The pH of the given soil sample was found to be **6.64**. This value falls in the slightly acidic range. At this pH, most essential plant nutrients remain reasonably available, although prolonged cultivation without liming may gradually lower the pH further and reduce the availability of nutrients such as phosphorus and molybdenum.

## Discussion

On commonly used classification scales, 6.64 sits at the boundary between “slightly acidic” (6.1–6.5) and “neutral” (6.6–7.3), close to the optimum range for most crops. This is consistent with the small lime requirement found in Experiment 3. Because the method relies on a single stabilised reading, the main error sources are incomplete equilibration of the suspension (reduced by the 30-minute stirring) and drift in electrode calibration, which should be checked periodically with standard buffers.

**In simple words:** The pH meter “feels” how many hydrogen ions are in the soil-water mix. More $H^+$ means more acidic. Our soil is almost neutral, which suits most crops.

# Experiment 2: Determination of the Electrical Conductivity (EC) of Soil

## Objective

To determine the electrical conductivity of a given soil sample in order to estimate the concentration of total soluble salts and assess the potential salinity hazard for crop production.

## Principle

Electrical conductivity (EC) is a measure of a solution's ability to conduct an electric current, which depends on the concentration of dissolved ions such as salts and minerals. It is typically expressed in microsiemens per centimetre (µS/cm). Pure water is a very poor conductor, whereas water containing dissolved salts conducts current approximately in proportion to the amount of salt present. Electrical conductance, the reciprocal of resistance, is therefore used for salinity appraisal.

*Example: EC = 0.000521 mhos/cm = 0.521 mS/cm = 521 µS/cm*

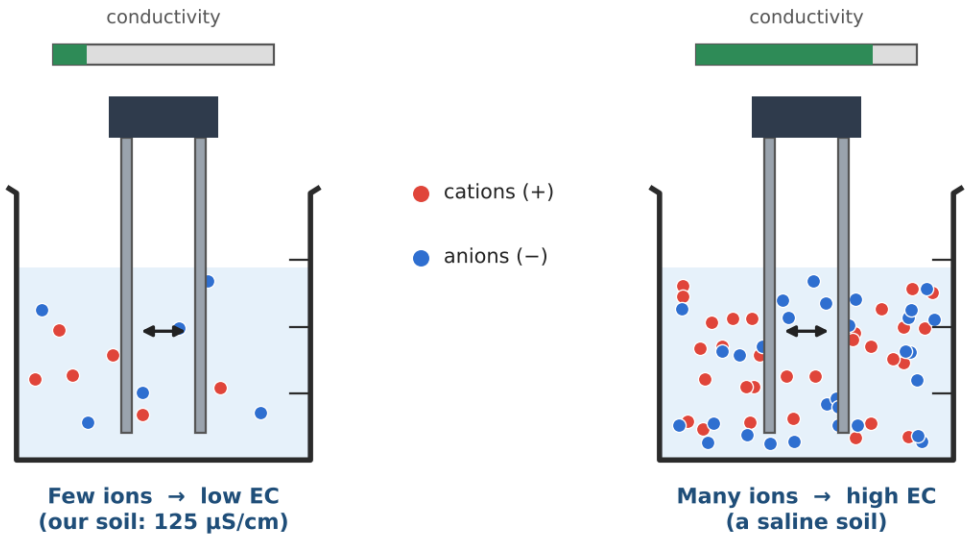



*Fig. 2.1: More dissolved ions carry more current, so the meter reads a higher EC.*

## Apparatus

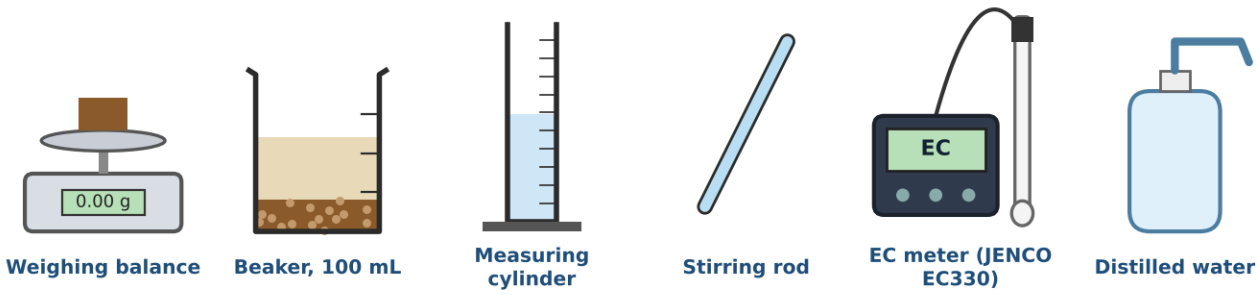



*Fig. 2.2: Apparatus used (schematic).*

## Procedure

1. 10 g of soil was weighed using a weighing balance.
2. The soil sample was poured into a 100 mL beaker.
3. 50 mL of distilled water was added to the sample.
4. The suspension was stirred for 30 minutes with a glass rod.
5. After 30 minutes, the EC meter electrode was placed in the beaker.
6. Once the reading had stabilised, the EC value was recorded.

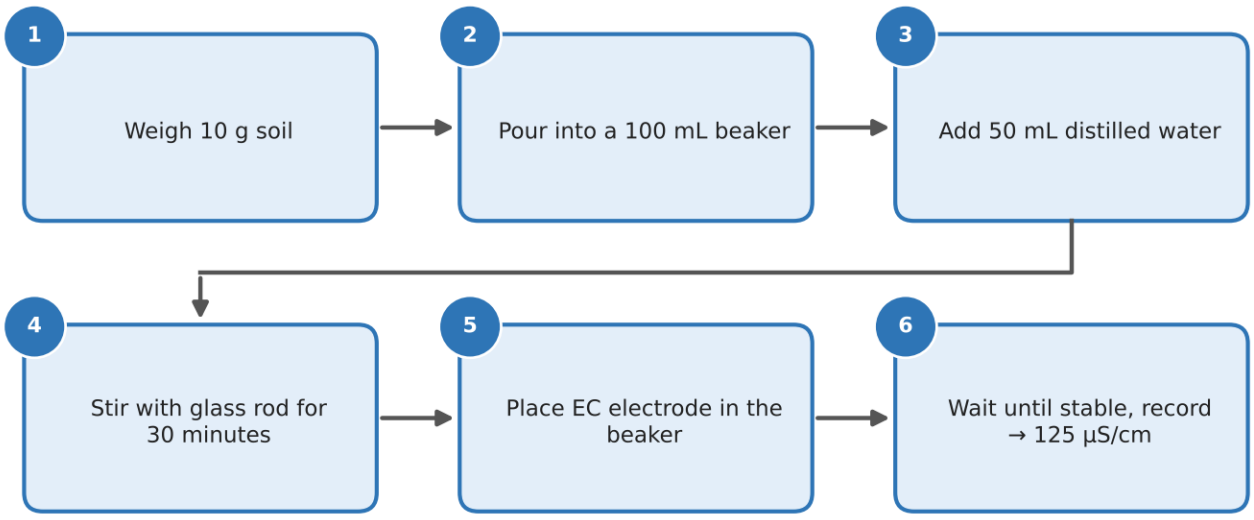



*Fig. 2.3: Procedure at a glance.*

## Observation

The stabilised reading obtained from the EC meter was: **EC = 125 µS/cm**

## Result

**Result:** The EC value of the soil sample was found to be **125 µS/cm**. This indicates a low soluble-salt content, which is generally favourable for plant growth. The soil is non-saline and is unlikely to cause any salinity stress to most crops.

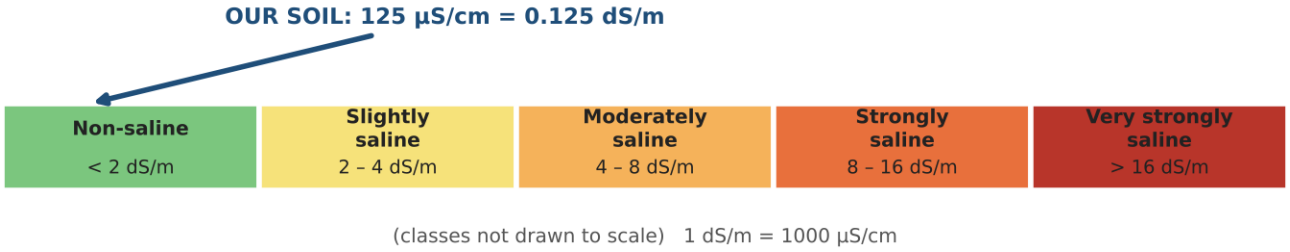



*Fig. 2.4: Where our soil sits on the salinity scale.*

## Discussion

In deciSiemens per metre, 125 µS/cm is about 0.125 dS/m, well below the threshold of roughly 2 dS/m used to define the lower boundary of “slightly saline” soils, so salinity is not expected to limit crop establishment or yield. Precautions: mix the suspension thoroughly for the full period and calibrate the EC probe against a standard KCl solution, since electrode fouling or drift can bias the reading.

**In simple words:** Salts dissolve into charged particles (ions) that carry electricity. Higher conductivity means saltier soil. At 125 µS/cm our soil is very low in salts.

# Experiment 3: Lime Requirement of Soil Sample

## Objective

To determine the amount of agricultural lime required to neutralise soil acidity and raise the pH of a given soil sample to a target optimum level for crop production.

## Principle

Plant growth is adversely affected by low soil pH. To bring the pH back towards neutrality, lime is added to the soil. Liming may be carried out using CaO, $Ca(OH)_2$, $CaCO_3$ (calcite) or $CaCO_3 \cdot MgCO_3$ (dolomite). Soil pH is the most important criterion determining the lime requirement, although texture, structure, organic-matter content, and the type and amount of clay also influence it. Lime neutralises soil acidity by reacting with $H^+$ ions in the soil solution, forming water and carbon dioxide and thereby raising the pH:

$$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

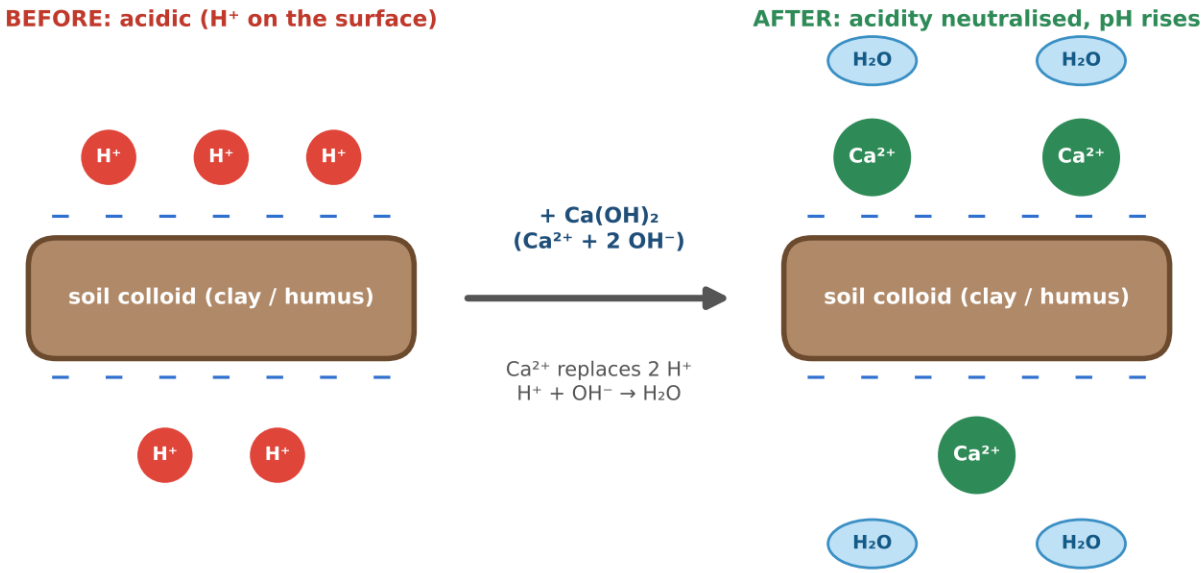



*Fig. 3.1: Calcium from the lime takes the place of $H^+$ on the soil surface, and $H^+ + OH^-$ forms water.*

## Apparatus and Reagent

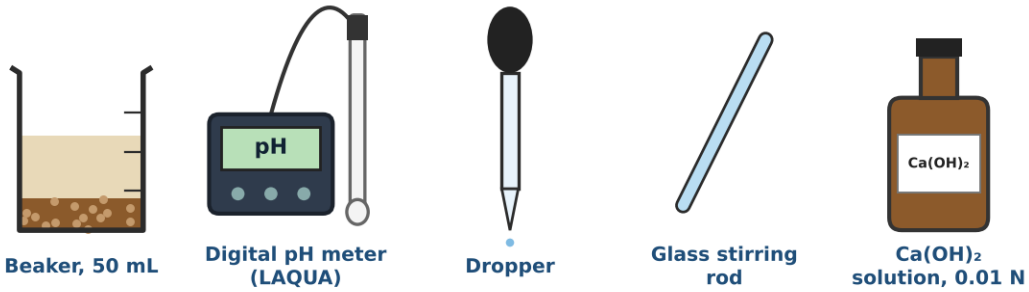



*Fig. 3.2: Apparatus and reagent ($Ca(OH)_2$ solution, 0.01 N).*

## Procedure

1. 10 g of soil was taken, 25 mL of distilled water was added, and the mixture was shaken for 30 minutes.
2. The pH of the resulting suspension was measured using the pH meter; this was recorded as the “0 drop” reading.
3. One drop of $Ca(OH)_2$ solution was added, the suspension was mixed, and the pH was recorded.
4. The addition of a single drop was repeated, with the pH recorded after each drop, up to a total of ten drops.

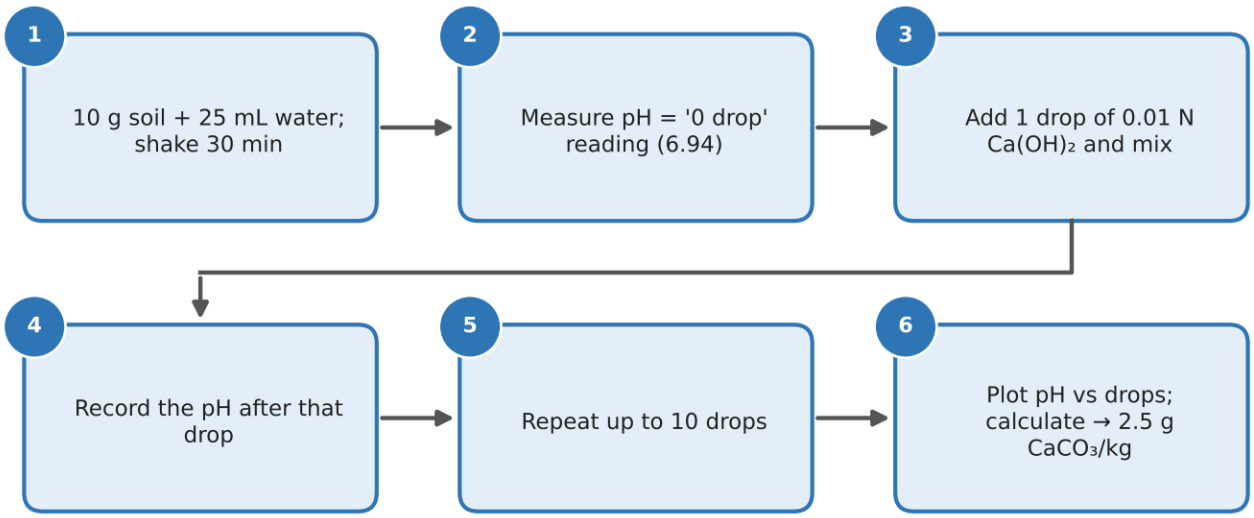



*Fig. 3.3: Procedure at a glance.*

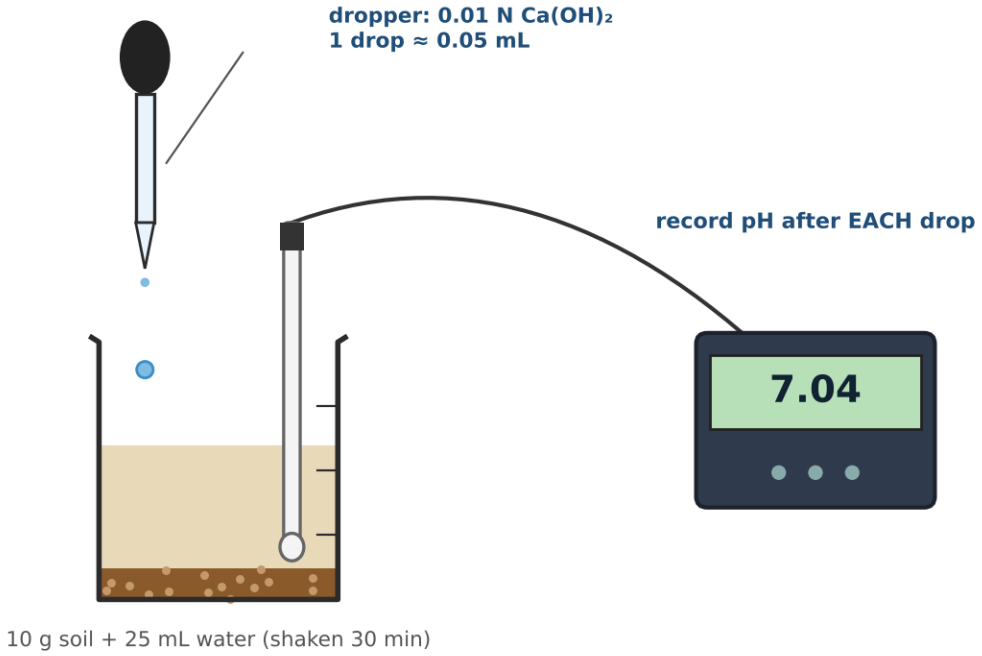



*Fig. 3.4: Adding Ca(OH)₂ one drop at a time while watching the pH.*

## Data Table

| Drops of Ca(OH)₂ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| pH | 6.94 | 7.04 | 7.13 | 7.32 | 7.48 | 7.58 | 7.64 | 7.67 | 8.04 | 8.13 | 8.19 |

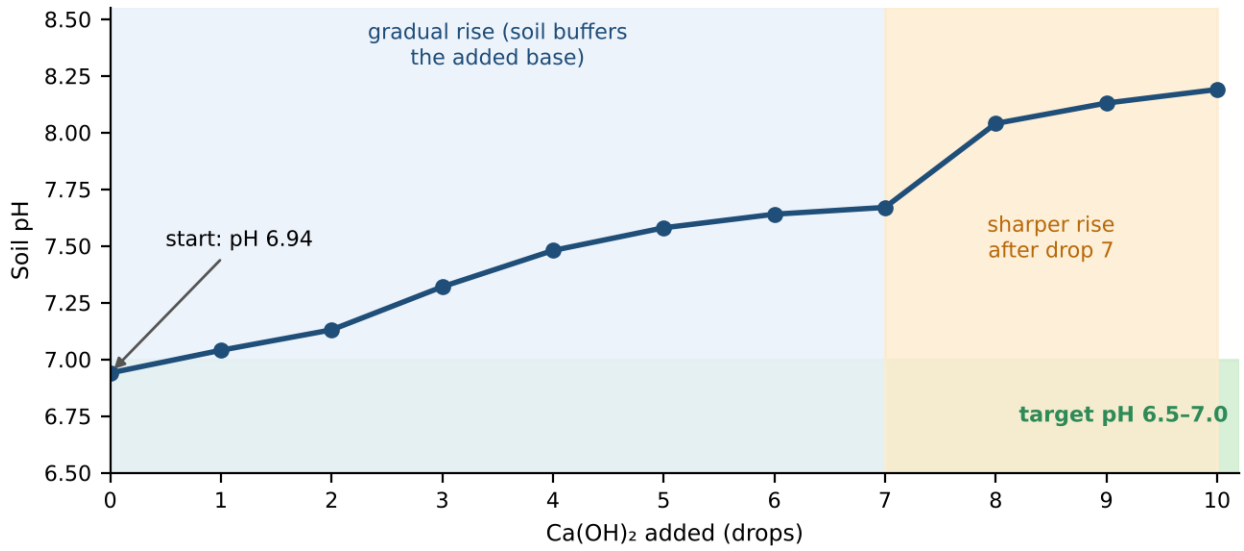



*Fig. 3.5: Change in soil pH with successive additions of $Ca(OH)_2$ (determination of lime requirement).*

## Calculation

One drop of $Ca(OH)_2$ is taken as ≈ 0.05 mL (standard drop volume).

$$\text{Total volume of } Ca(OH)_2 \text{ added (10 drops)} = 10 \times 0.05 = 0.5 \text{ mL}$$

Since 10 g of soil was used, the scaling factor to express the result per kg of soil is:

$$\text{Scaling factor} = 1000 / 10 = 100$$

$$\text{Lime requirement (g } CaCO_3 \text{ per kg soil)} = [\ (V \times N \times 50) / \text{Weight of soil (g)}\ ] \times \text{Scaling factor}$$

Here V = 0.5 mL, N = 0.01 N, weight of soil = 10 g, scaling factor = 100 (50 is the equivalent-weight conversion factor for $CaCO_3$):

$$\text{Lime requirement} = [\ (0.5 \times 0.01 \times 50) / 10\ ] \times 100 = 2.5 \text{ g } CaCO_3 \text{ per kg of soil}$$

## Result

**Result:** The lime requirement of the given soil sample was found to be **2.5 g of $CaCO_3$ per kg of soil**. The soil pH rose steadily from 6.94 to 8.19 with successive additions of $Ca(OH)_2$, with a comparatively sharper rise beyond the 7th drop, indicating the buffering capacity of the soil against the applied base.

## Discussion

The curve rises gradually up to about the 7th drop and then more steeply: early base is consumed neutralising readily exchangeable acidity, after which small additions raise the pH more. Because the target pH for most crops is close to neutral (6.5–7.0) and the initial pH (6.94) was already near it, only a modest lime application was calculated. Main errors: the assumed fixed drop volume (0.05 mL) and electrode drift; both can be reduced by recalibrating the meter and using a calibrated micro-burette.

**In simple words:** The soil's acidity is like a bank account of $H^+$ ions. We add a known amount of lime-water and watch the pH climb. How much we had to add tells us how much lime a whole kilogram of soil would need.

# Experiment 4: Determination of Organic Carbon of Soil Sample

*Walkley and Black Wet Oxidation Method*

## Objective

To determine the percentage of organic carbon present in a soil sample by the Walkley and Black wet-oxidation method.

## Principle

Soil organic carbon is oxidised by potassium dichromate ($K_2Cr_2O_7$) in the presence of concentrated sulfuric acid ($H_2SO_4$); the heat generated on dilution of the acid assists the oxidation. Organic-carbon compounds act as strong reducing agents, and this reducing capacity is exploited to determine organic carbon by digestion with a standard oxidising agent, chromic acid:

$$4Cr^{6+} + 3C \rightarrow 4Cr^{3+} + 3C^{4+}$$

$$K_2Cr_2O_7 + 3C + 6H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 3CO_2 + 8H_2O + H_2Cr_2O_7 \text{ (unreduced)}$$

The excess (unreduced) chromic acid remaining after oxidation is back-titrated with standard ferrous sulfate solution:

$$K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$$

and the amount of carbon oxidised is calculated from the quantity of chromic acid reduced.

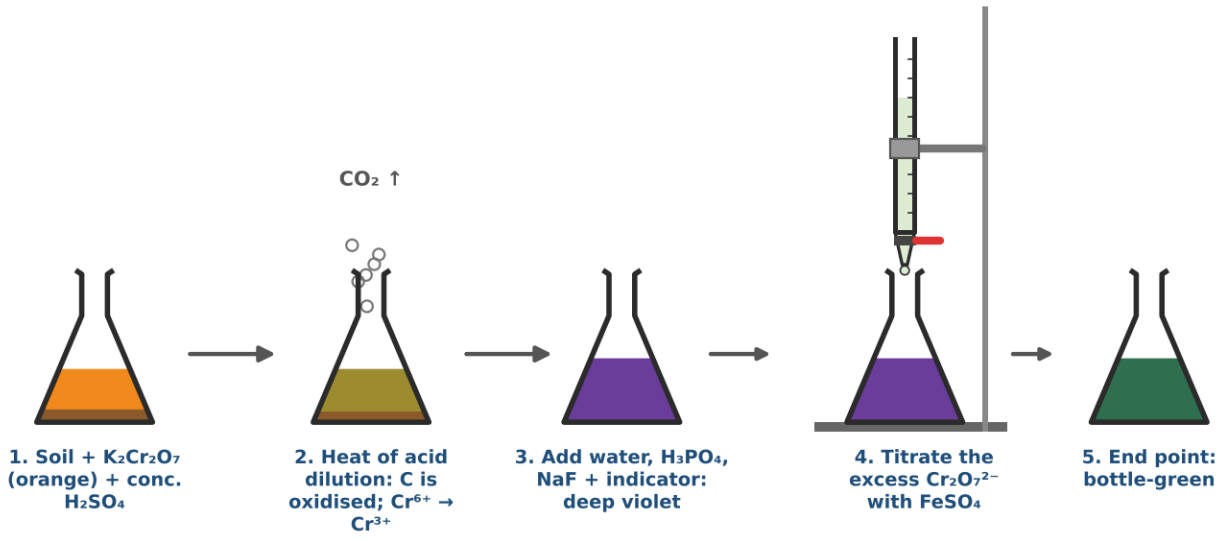



*Fig. 4.1: The story of the experiment: soil carbon uses up part of the dichromate; $FeSO_4$ measures the rest.*

## Apparatus and Reagents

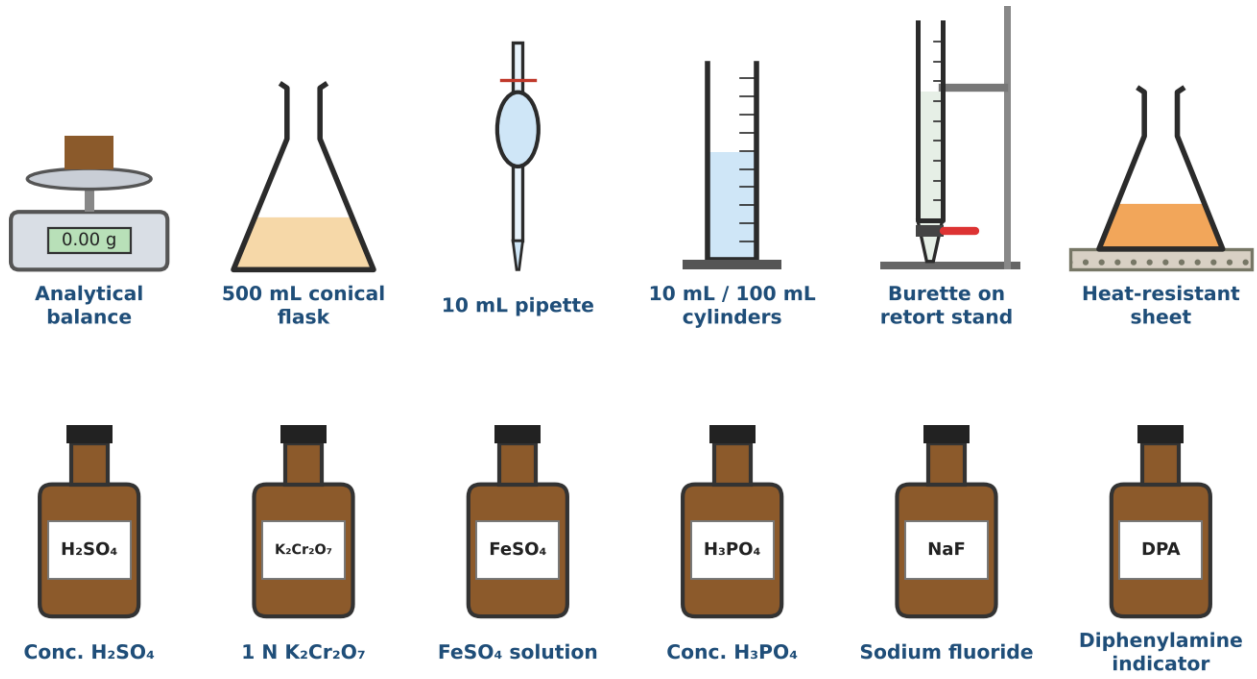



*Fig. 4.2: Apparatus (top row) and reagents (bottom rows). Standard 1 N $K_2Cr_2O_7$ = 49.03 g/L; $FeSO_4$ = 278 g $FeSO_4·7H_2O$ + 15 mL conc. $H_2SO_4$ per litre.*

## Procedure

1. 2 g of air-dried soil (passed through a 2 mm sieve) was transferred to a 500 mL clean, dry conical flask.
2. 10 mL of standard 1 N $K_2Cr_2O_7$ solution was added using a pipette.
3. 10 mL of concentrated $H_2SO_4$ was added with a measuring cylinder and mixed thoroughly.
4. The flask was allowed to cool on a heat-resistant sheet with occasional shaking for 30 minutes.
5. After cooling, about 150 mL of distilled water, 10 mL of phosphoric acid and 0.2 g of sodium fluoride were added.
6. 3 mL (about 60 drops) of diphenylamine indicator was added, producing a deep violet colour.
7. The excess chromic acid was titrated with standard ferrous sulfate solution until the colour changed sharply to deep bottle-green (end point).
8. The titration was repeated three times, and a blank was run in the same manner, omitting the soil, to standardise the $FeSO_4$ solution.

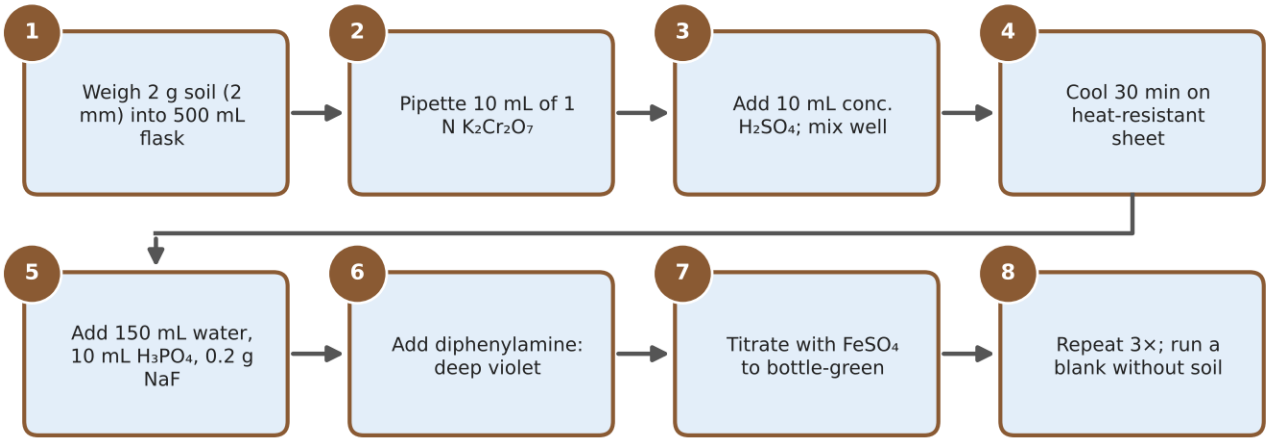



*Fig. 4.3: Procedure at a glance.*

## Data Table

| Observation | Initial burette reading (mL) | Final burette reading (mL) | Difference (mL) | Mean (mL) |
|---|---|---|---|---|
| Soil sample (T) – 1 | 10 | 25 | 15.0 | 14.65 |
| Soil sample (T) – 2 | 25 | 39.3 | 14.3 | |
| Blank (B) – 1 | 0 | 20.6 | 20.6 | 20.45 |
| Blank (B) – 2 | 20.6 | 40.9 | 20.3 | |

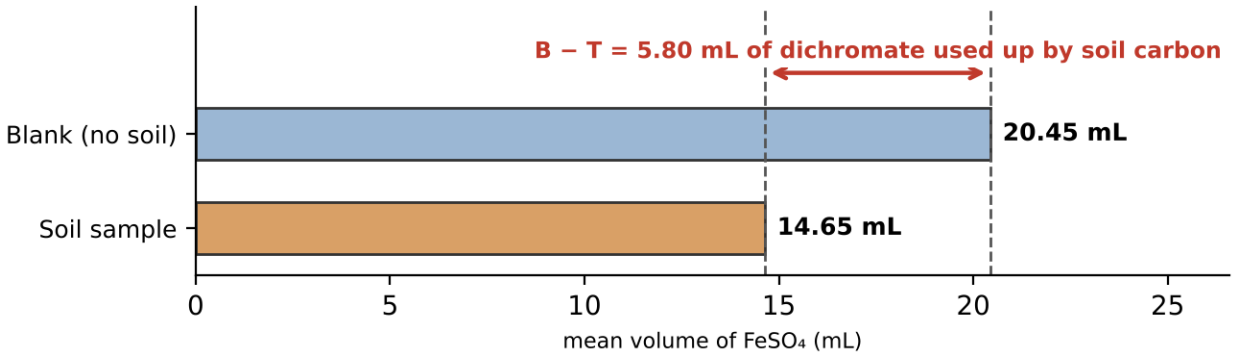



*Fig. 4.4: Blank vs soil titre. The soil needs less $FeSO_4$ because soil carbon already used up some dichromate.*

## Standardisation of $FeSO_4$ Solution

Using $S_1V_1 = S_2V_2$ with $S_1 = 1$ N, $V_1 = 10$ mL and $V_2$ = mean blank volume = 20.45 mL:

$$S_2 = (S_1 \times V_1) / V_2 = (1 \times 10) / 20.45 = 0.489 \text{ N}$$

## Calculation

1000 mL of 1 N $K_2Cr_2O_7$ is equivalent to 3 g of carbon (equivalent weight of C = 12/4 = 3), so 1 mL corresponds to 0.003 g C. Only about 77 % of soil carbon is oxidised under these conditions, so the factor 0.003 is multiplied by 1.3:

$$\% \text{ Organic carbon} = [\,(B - T) \times f \times 0.003 \times 1.3 \times 100\,] / W$$

B = 20.45, T = 14.65, f = 0.489 (strength of $FeSO_4$), W = 2 g:

$$\% \text{ Organic carbon} = [\,(20.45 - 14.65) \times 0.489 \times 0.003 \times 1.3 \times 100\,] / 2 = 0.55\ \%$$

## Result

**Result:** The organic carbon content of the given soil sample was found to be **0.55 %**. Every 100 g of soil contains approximately 0.55 g of organic carbon, indicating a moderate level of organic matter.

## Discussion

Multiplying by the Van Bemmelen factor of 1.724 gives soil organic matter of about 0.55 × 1.724 ≈ 0.95 %, on the lower side for agricultural soils, suggesting compost or crop-residue incorporation. The duplicate titres for soil (15.0 and 14.3 mL) and blank (20.6 and 20.3 mL) are reasonably close. Because Walkley–Black oxidises only about 77 % of the carbon (hence the 1.3 factor), the result is an estimate; incomplete mixing and subjective judgement of the end-point colour are the main error sources.

**In simple words:** Think of the dichromate as a fixed amount of “carbon-eating” chemical. Soil carbon eats part of it; we then measure the leftover with $FeSO_4$. The less is left, the more carbon the soil had.

# Experiment 5: Determination of Total Nitrogen in Soil Sample

*Kjeldahl Method*

## Objective

To determine the percentage of total nitrogen present in a given soil sample by the Kjeldahl method.

## Principle

The Kjeldahl procedure involves three steps. **Digestion:** organic nitrogen is converted into ammonium sulfate by heating with concentrated sulfuric acid and a catalyst. **Distillation:** the digest is made alkaline with sodium hydroxide, liberating ammonia gas. **Titration:** the ammonia is absorbed in boric acid and titrated against a standard acid.

$$\text{Soil–N} + H_2SO_4 \xrightarrow{\text{(heat, 360–410 °C)}} (NH_4)_2SO_4$$

$$(NH_4)_2SO_4 + 2NaOH \rightarrow NH_3\uparrow + H_2O + Na_2SO_4$$

$$NH_3 + H_3BO_3 \rightarrow NH_4H_2BO_3 \text{ (ammonium borate)}$$

$$NH_4H_2BO_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4 + H_3BO_3$$

The digestion temperature is critical: below 360 °C the release of ammonia is slow and incomplete, while above 410 °C some ammonia is lost from the mixture.

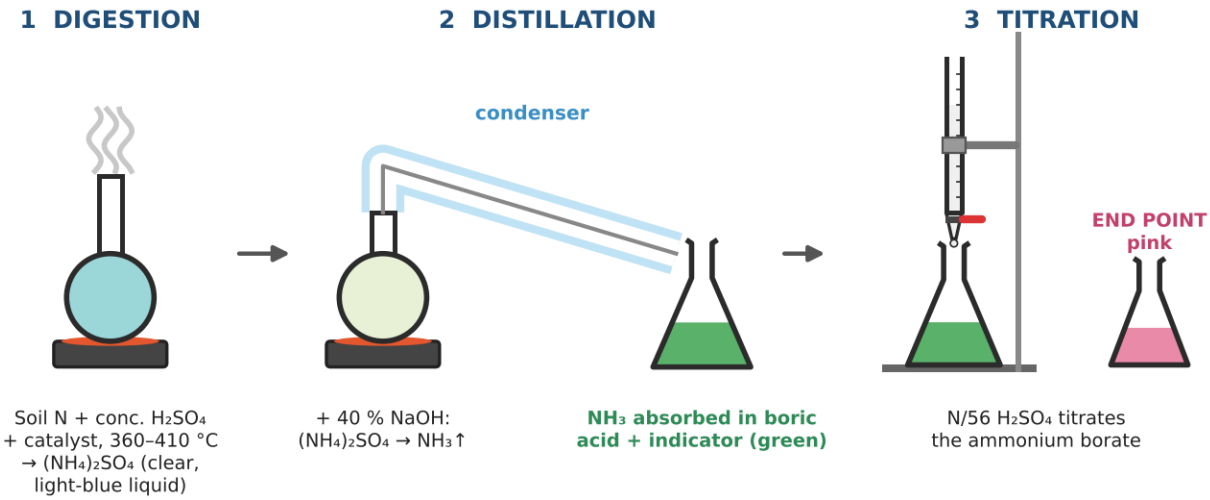



*Fig. 5.1: The three Kjeldahl stages: digestion, distillation, titration.*

## Apparatus and Reagents

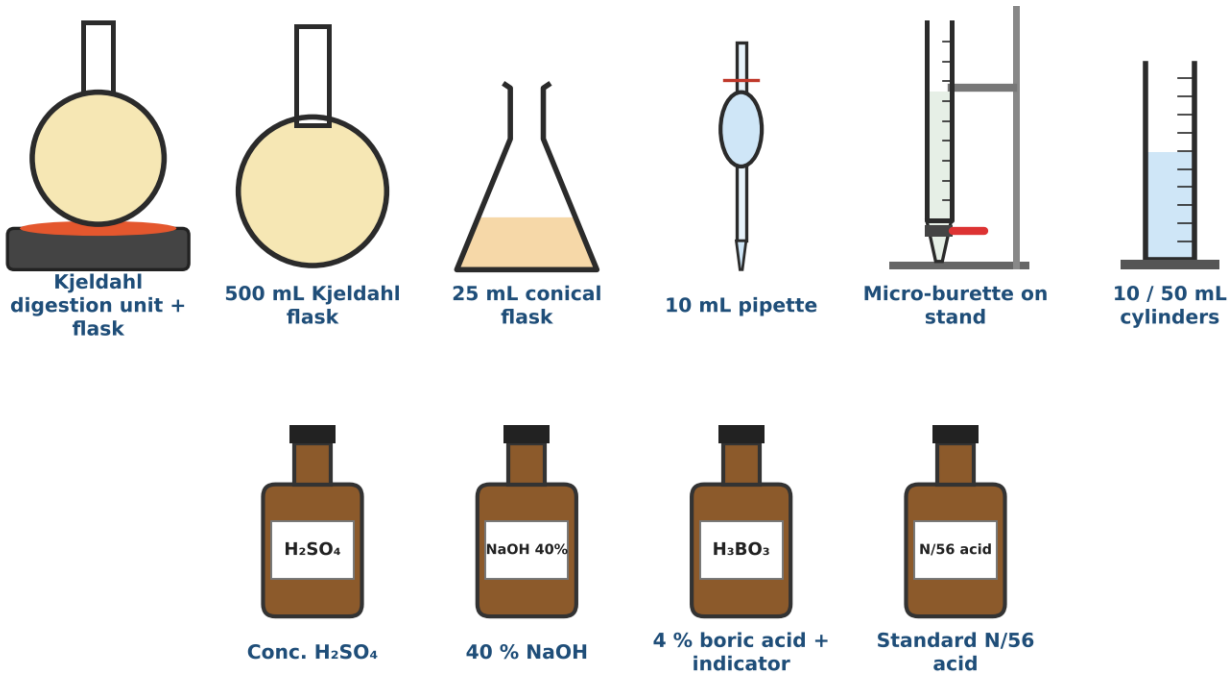



*Fig. 5.2: Apparatus (top row) and reagents (bottom row). Digestion mixture: $CuSO_4 \cdot 5H_2O$, selenium powder and anhydrous $Na_2SO_4$.*

## Procedure

1. 5 g of finely powdered soil was placed in a 500 mL Kjeldahl flask, 20 mL of distilled water was added, and the mixture was left for 20 minutes.
2. 25 mL of concentrated $H_2SO_4$ was added and mixed, and the flask was heated over a low flame for 15 minutes.
3. When white fumes of $H_2SO_4$ appeared, the flask was removed and 2–3 g of catalyst mixture was added.
4. Digestion was continued with slowly rising temperature for two hours or more until the liquid became clear (light bluish colour).
5. The digest was cooled, diluted with 100 mL distilled water, and made up to volume in a 250 mL volumetric flask.
6. An aliquot of the extract was distilled with 40 % NaOH in the micro-Kjeldahl apparatus, and the liberated ammonia was collected in 10 mL of 4 % boric acid containing mixed indicator.
7. The distillate was titrated against standard $H_2SO_4$ until the colour changed to pink (end point).
8. A blank determination was carried out simultaneously using all reagents but no soil.

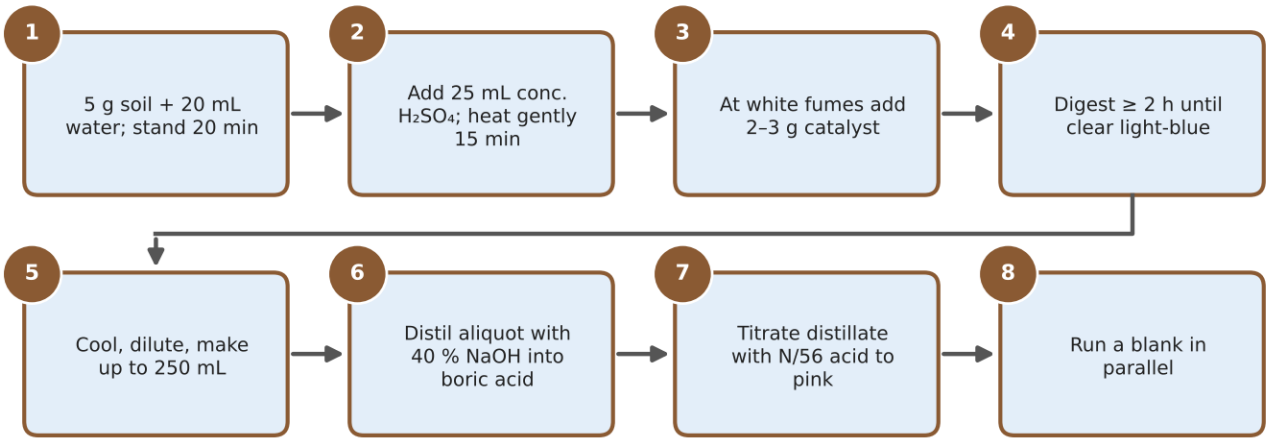



*Fig. 5.3: Procedure at a glance.*

## Data Table

| Observation | Initial burette reading (mL) | Final burette reading (mL) | Difference (mL) | Average (mL) |
|---|---|---|---|---|
| Soil sample (T) – 1 | 1.1 | 3.1 | 2.0 | 1.8 |
| Soil sample (T) – 2 | 3.1 | 4.7 | 1.6 | |
| Blank (B) – 1 | 6.7 | 8.2 | 1.5 | 1.2 |
| Blank (B) – 2 | 8.2 | 9.1 | 0.9 | |

T = 1.8 mL, B = 1.2 mL, f (normality factor of standard acid) = 0.018, w (weight of soil) = 5 g, volume of extract used = 10 mL.

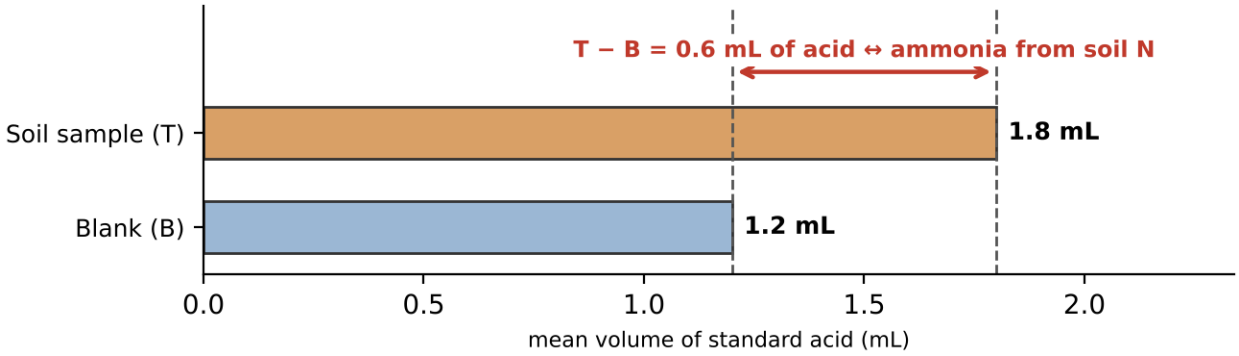



*Fig. 5.4: The extra acid needed by the soil sample over the blank corresponds to the soil nitrogen.*

## Calculation

1000 mL of 1 N $H_2SO_4$ is equivalent to 14 g of N, so 1 mL of 1 N $H_2SO_4$ corresponds to 0.014 g N.

$$\% \text{ N in soil} = [\,(T - B) \times f \times 0.014 \times 250 \times 100\,] / (w \times \text{volume of extract used})$$

$$\% \text{ N in soil} = [\,(1.8 - 1.2) \times 0.018 \times 0.014 \times 250 \times 100\,] / (5 \times 10) = 0.0756\ \%$$

## Result

**Result:** The total nitrogen content of the given soil sample was found to be **0.0756 %**, equivalent to 0.756 g N per kg of soil.

## Discussion

0.756 g N/kg is in the low range typical of many cultivated tropical and sub-tropical soils. Comparing with the organic carbon from Experiment 4 gives C : N ≈ 0.55 / 0.0756 ≈ 7.3 : 1, narrower than the 10–12 : 1 usually reported for surface soils; this may indicate rapid mineralisation or simply combined analytical uncertainty and would be worth repeating. The blank duplicates (1.5 and 0.9 mL) show a wider spread than the soil duplicates (2.0 and 1.6 mL), suggesting the blank end point was harder to judge. Error sources: ammonia loss if the delivery tube is not sealed below the boric acid surface, incomplete digestion, and indicator subjectivity.

**In simple words:** Digestion turns all the soil's nitrogen into ammonium. Adding strong alkali turns it into ammonia gas, which we catch in boric acid and then measure with acid. The amount of acid used is proportional to the nitrogen.

# Experiment 6: Determination of Calcium ($Ca^{2+}$) and Magnesium ($Mg^{2+}$) by EDTA Titration

## Objective

To determine the concentration of exchangeable $Ca^{2+}$ and $Mg^{2+}$ ions in a given soil sample by complexometric titration with EDTA.

## Principle

Calcium and magnesium in the soil are extracted using 1 N ammonium acetate ($NH_4OAc$) solution and determined by titration with ethylenediaminetetraacetic acid (EDTA), which forms stable, soluble 1 : 1 complexes with both ions:

$$Ca^{2+} + EDTA^{4-} \rightarrow [Ca\text{–}EDTA]^{2-}$$

$$Mg^{2+} + EDTA^{4-} \rightarrow [Mg\text{–}EDTA]^{2-}$$

For calcium alone, murexide indicator is used at about pH 11–12; the colour changes from pink to light blue at the end point (magnesium is masked as $Mg(OH)_2$ precipitate). For combined calcium and magnesium, Eriochrome Black-T (EBT) is used at about pH 10. Magnesium is then obtained by subtracting the calcium value from the combined (Ca + Mg) value.

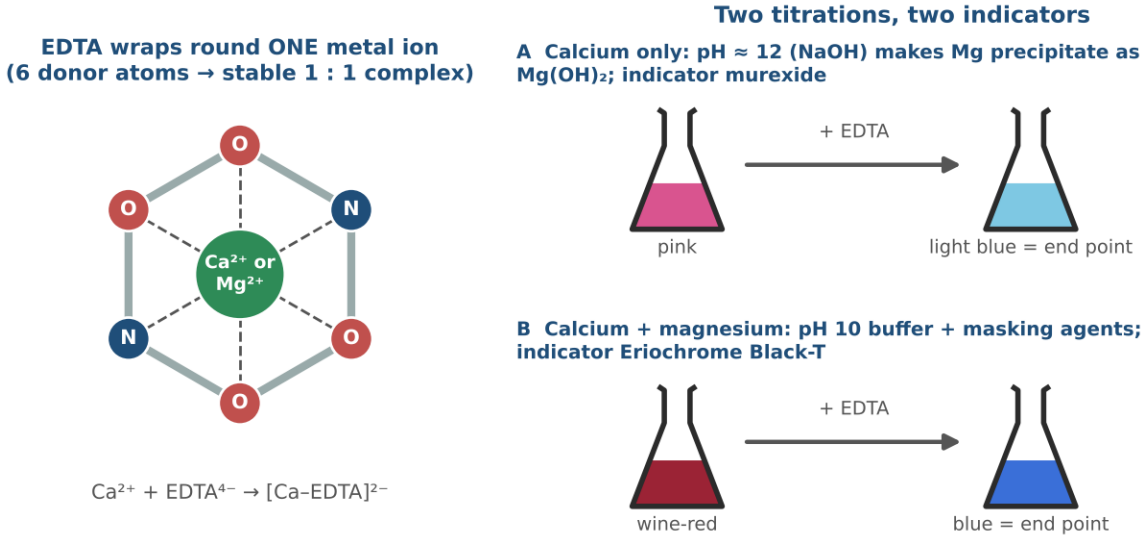



*Fig. 6.1: How EDTA grabs a metal ion, and the two titrations used.*

## Apparatus and Reagents

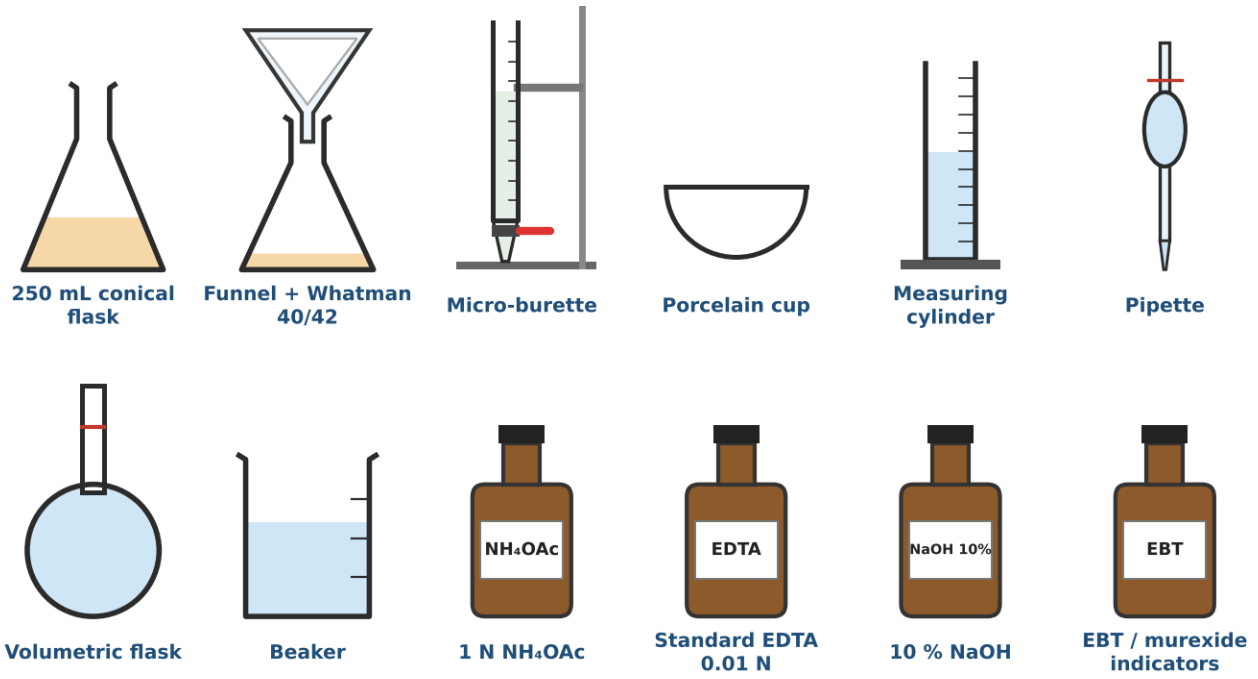



*Fig. 6.2: Apparatus (top) and reagents (bottom). Also used: pH 10 buffer ($NH_4Cl$ + $NH_4OH$), hydroxylamine hydrochloride, potassium ferrocyanide, triethanolamine.*

## Procedure

**A. Extraction of Ca and Mg**

1. 10 g of air-dried soil was taken in a 250 mL conical flask and 100 mL of neutral 1 N $NH_4OAc$ solution was added.
2. The flask was shaken thoroughly and allowed to stand overnight, then filtered through Whatman No. 40/42 paper to obtain the soil extract.

**B. Determination of calcium**

1. 2 mL of soil extract was taken in a porcelain cup and 5 mL of 10 % NaOH solution was added to precipitate $Mg(OH)_2$.
2. About 0.02 g of murexide indicator was added, and the solution was titrated with 0.01 N EDTA until the colour changed from pink to light blue (end point).
3. A blank was run using distilled water in place of soil extract, and the EDTA was standardised against a standard calcium solution.

**C. Determination of calcium + magnesium**

1. 2 mL of soil extract was taken in a porcelain cup with 25 mL distilled water, 10–15 mL of pH 10 buffer, and a few drops each of potassium ferrocyanide, hydroxylamine hydrochloride and TEA solutions.
2. The solution was warmed gently, cooled, and 3 drops of Eriochrome Black-T indicator were added.
3. The solution was titrated with EDTA until the colour changed at the end point, and a corresponding blank was run.

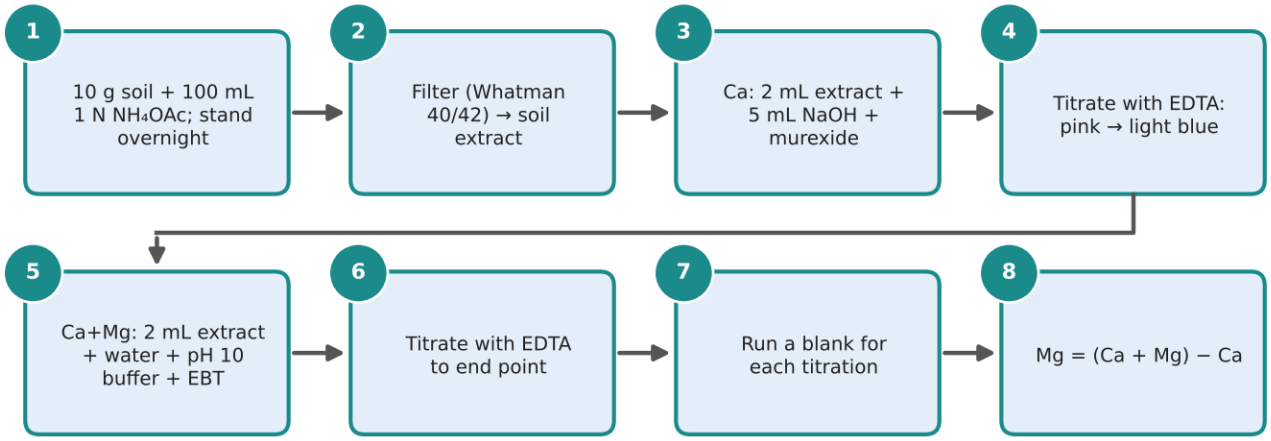



*Fig. 6.3: Procedure at a glance.*

## Observations

**Calcium determination**

| Observation | Initial (mL) | Final (mL) | Difference (mL) | Mean (mL) |
|---|---|---|---|---|
| Soil sample – 1 | 0 | 5.2 | 5.2 | 5.2 |
| Soil sample – 2 | 5.2 | 10.4 | 5.2 | |
| Blank | 33.5 | 34.5 | 1.0 | 1.0 |

Corrected Ca reading = 5.2 − 1.0 = 4.2 mL

**Calcium + magnesium determination**

| Observation | Initial (mL) | Final (mL) | Difference (mL) | Mean (mL) |
|---|---|---|---|---|
| Soil sample – 1 | 0 | 8.2 | 8.2 | 8.1 |
| Soil sample – 2 | 8.2 | 16.2 | 8.0 | |
| Blank | 0 | 1.0 | 1.0 | 1.0 |

Corrected (Ca + Mg) reading = 8.1 − 1.0 = 7.1 mL; Mg reading = 7.1 − 4.2 = 2.9 mL

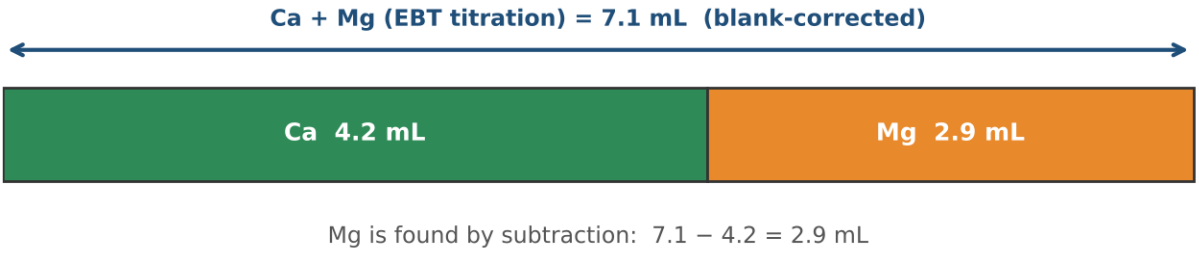



*Fig. 6.4: The titre of the Ca + Mg titration is split into its calcium and magnesium parts.*

## Calculation

Calcium: 1 mL of 1 N EDTA ≡ 0.02 g Ca.

*% Ca = (0.02 × mL EDTA × N × total extract volume × 100) / (mL extract used × weight of soil)*

With EDTA = 4.2 mL, N = 0.0048, extract used = 2 mL, soil = 10 g, total extract = 100 mL:

*% Ca = (0.02 × 4.2 × 0.0048 × 100 × 100) / (2 × 10) = 0.20 % (= 2000 ppm)*

Magnesium: 1 mL of 1 N EDTA ≡ 0.012 g Mg; with EDTA = 2.9 mL:

*% Mg = (0.012 × 2.9 × 0.0048 × 100 × 100) / (2 × 10) = 0.0835 % (= 835 ppm)*

## Result

| Nutrient | Percentage (%) | Concentration (ppm) |
|---|---|---|
| Calcium (Ca) | 0.20 | 2000 |
| Magnesium (Mg) | 0.0835 | 835 |

**Result:** The available calcium and magnesium contents were found to be **0.20 % (2000 ppm)** and **0.0835 % (835 ppm)**, respectively, indicating an adequate base status for most crops.

## Discussion

The Ca : Mg ratio is about 2000 / 835 ≈ 2.4 : 1. A ratio of about 3–6 : 1 is often considered favourable, so a lower ratio may indicate a comparatively high share of magnesium, which can occasionally interfere with potassium and calcium uptake in sensitive crops. The duplicate titres agree well (5.2 and 5.2 mL; 8.2 and 8.0 mL). Error sources: subjective colour end points, incomplete masking of interfering ions and loss of extract during filtration; titrate slowly near the end point and use fresh indicators.

**In simple words:** EDTA is a molecular “claw” that grabs calcium and magnesium one-to-one. When all the metal is grabbed, the indicator changes colour. By doing one titration for Ca alone and one for Ca + Mg, we get Mg by subtraction.

# Experiment 7: Determination of Chloride ($Cl^-$) in Soil Sample

*Mohr's Argentometric Titration Method*

## Objective

To determine the amount of chloride present in a soil sample by extracting soluble chloride with distilled water and titrating it against standard silver nitrate ($AgNO_3$), using potassium chromate ($K_2CrO_4$) as the indicator.

## Principle

Chloride ions in the soil extract react with silver ions from $AgNO_3$ to form a white precipitate of silver chloride:

$$Ag^+ + Cl^- \rightarrow AgCl\downarrow$$

Once all the chloride has precipitated, the first excess of silver ions reacts with the chromate ions of the indicator to form brick-red silver chromate, marking the end point:

$$2Ag^+ + CrO_4^{2-} \rightarrow Ag_2CrO_4\downarrow \text{ (brick-red)}$$

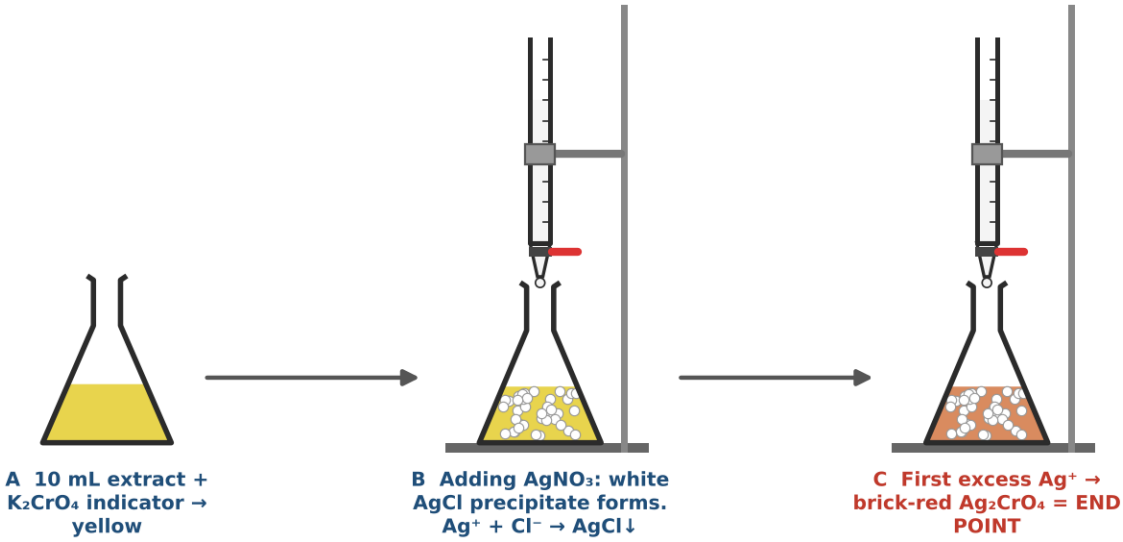



*Fig. 7.1: The colour story of Mohr's titration: yellow, white cloudiness, then the first brick-red tint.*

## Apparatus and Reagents

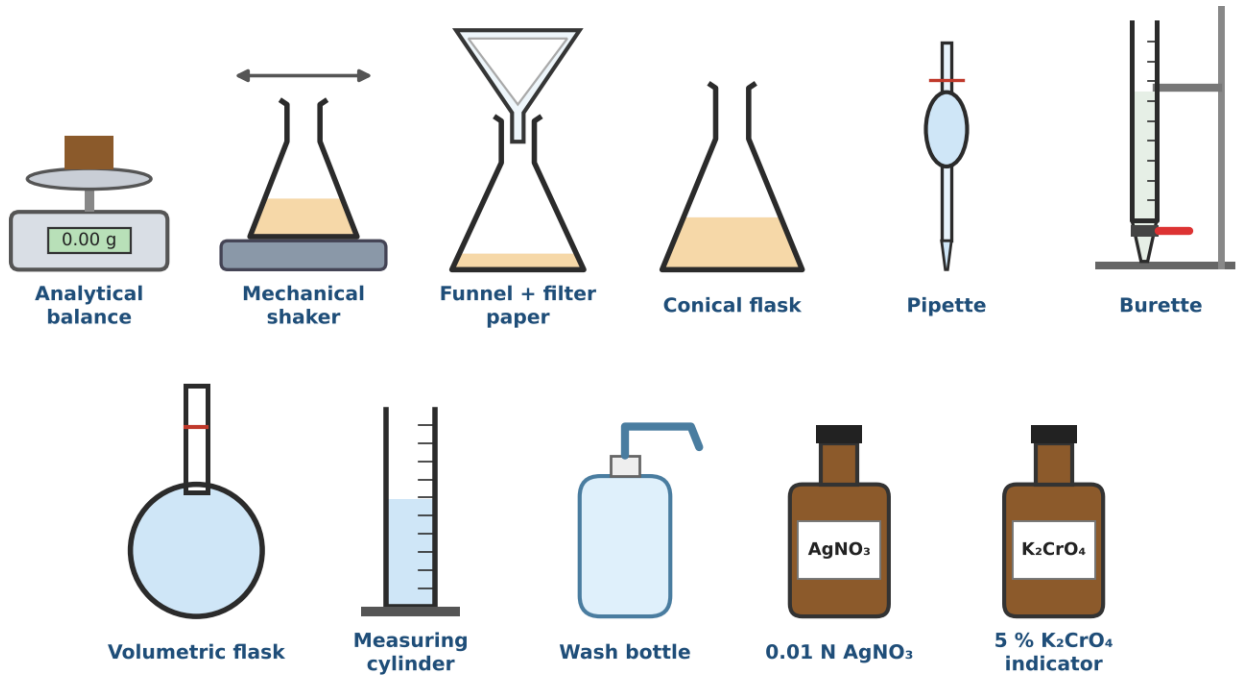



*Fig. 7.2: Apparatus and reagents (0.01 N $AgNO_3$; 5 % $K_2CrO_4$ indicator).*

## Procedure

**Preparation of soil extract**

1. 10 g of air-dried, sieved soil was weighed into a 250 mL conical flask, and 50 mL of distilled water was added (soil : water = 1 : 5).
2. The mixture was shaken for 30 minutes, allowed to settle, and filtered through Whatman No. 42 filter paper to obtain the clear soil extract.

**Titration**

1. For the blank, 10 mL of distilled water was pipetted into a conical flask, 1 mL of 5 % $K_2CrO_4$ was added, and the mixture was titrated with $AgNO_3$ until a permanent faint brick-red colour appeared.
2. For the sample, 10 mL of soil extract was pipetted into a clean flask, 1 mL of indicator was added (solution turned yellow), and the mixture was titrated with 0.01 N $AgNO_3$ with continuous swirling until the first permanent brick-red/reddish-brown colour appeared.
3. The titration was repeated to obtain concordant readings.

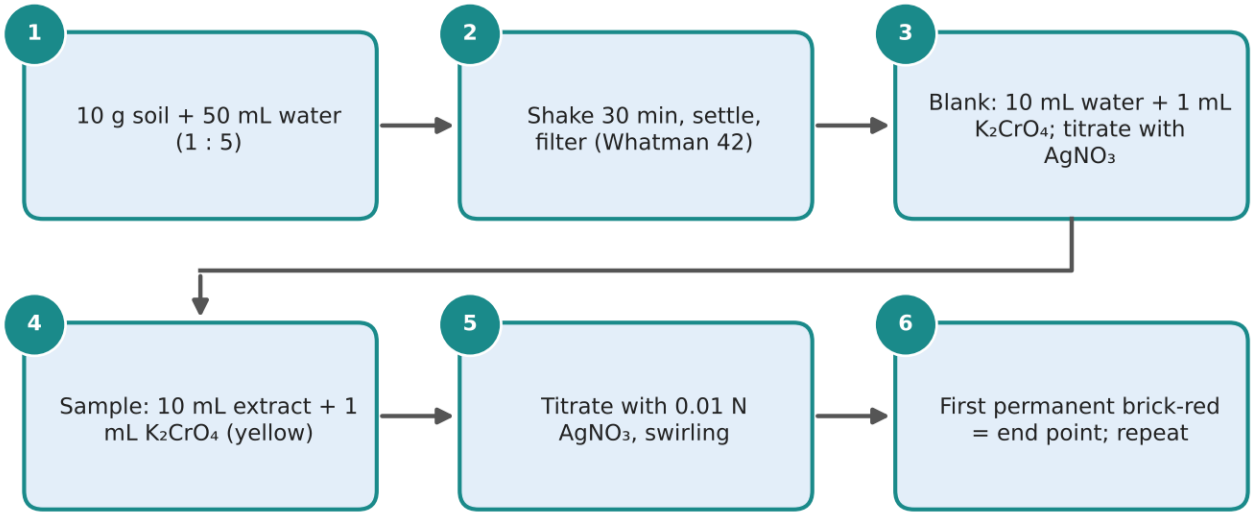



*Fig. 7.3: Procedure at a glance.*

## Data Table

| Observation | Initial (mL) | Final (mL) | Difference (mL) | Mean (mL) |
|---|---|---|---|---|
| Soil sample – 1 | 0 | 3.5 | 3.5 | 3.5 |
| Soil sample – 2 | 3.5 | 7.0 | 3.5 | |
| Blank | 7.0 | 8.8 | 1.8 | 1.8 |

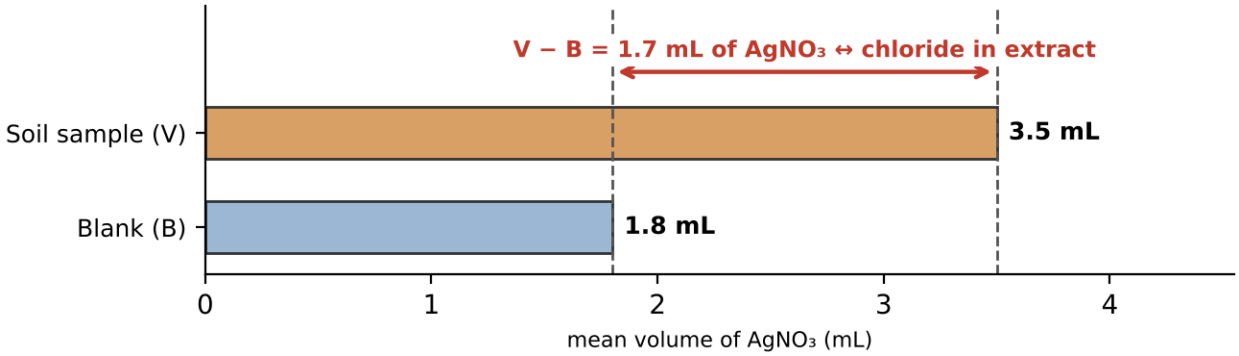



*Fig. 7.4: Sample vs blank titre; the difference is the silver used by chloride.*

## Calculation

Weight of soil = 10 g, total extract = 50 mL, extract titrated $V_s$ = 10 mL, N($AgNO_3$) = 0.01 N, equivalent weight of $Cl^-$ = 35.45 g/eq, V = 3.5 mL, B = 1.8 mL.

$$Cl^- \text{ (mg/L)} = [\,(V - B) \times N \times 35.45 \times 1000\,] / V_s = [\,(3.5 - 1.8) \times 0.01 \times 35.45 \times 1000\,] / 10 = 60.265 \text{ mg/L}$$

Accounting for the soil-to-water extraction ratio (10 g soil in 50 mL water):

$$Cl^- \text{ (mg/kg)} = 60.265 \times (0.05 / 0.01) = 301.325 \text{ mg/kg}$$

## Result

**Result:** The available chloride ($Cl^-$) content of the soil sample was found to be **301.325 mg/kg (≈ 301 ppm)**, a low-to-moderate level with no apparent chloride toxicity; this level is not normally harmful to most crops.

## Discussion

The two replicate titrations gave identical titres (3.5 mL each), indicating good reproducibility. Chloride toxicity in sensitive crops is generally associated with levels well above 350–700 mg/kg (thresholds vary with crop, soil and irrigation water), so about 301 mg/kg is unlikely to be harmful, though it should be monitored where chloride-sensitive crops or saline irrigation water are involved. Main errors: over-titration past the end point (the pale brick-red is subtle against the yellow) and interference from other anions; titrate slowly with swirling and compare with the blank colour.

**In simple words:** Silver grabs chloride first and makes a white cloud. Only when all the chloride is gone does silver react with the yellow chromate indicator and turn the solution red-brown. The red tint tells us we are done.

# Experiment 8: Determination of Potassium ($K^+$) in Soil by Flame Photometer

## Objective

To determine the amount of available potassium ($K^+$) present in a soil sample using a flame photometer.

## Principle

When a solution containing potassium ions is introduced into the flame of a flame photometer, the potassium atoms are excited by the heat. As the excited atoms return to the ground state they emit light of a characteristic wavelength, for potassium about 766 nm. The intensity of the emitted light is proportional to the potassium concentration within the working range of the instrument, which is found by comparing the extract with standard potassium solutions.

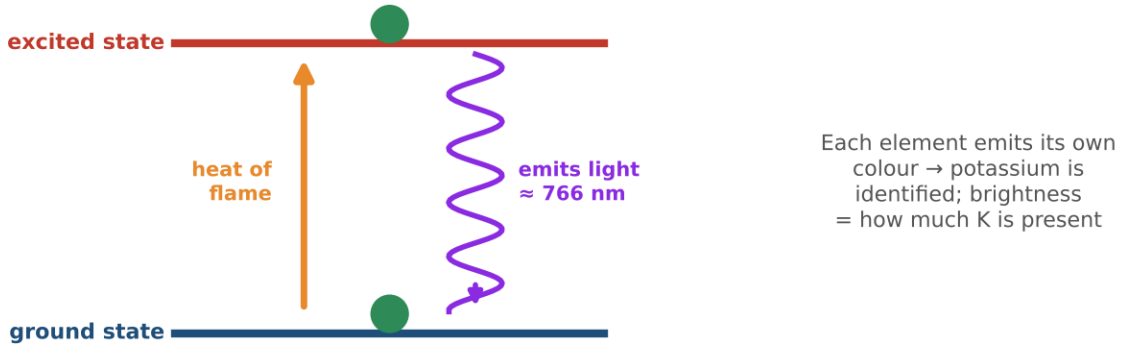



*Fig. 8.1: Why a flame can "read" potassium: excited atoms give off light of a fixed wavelength.*

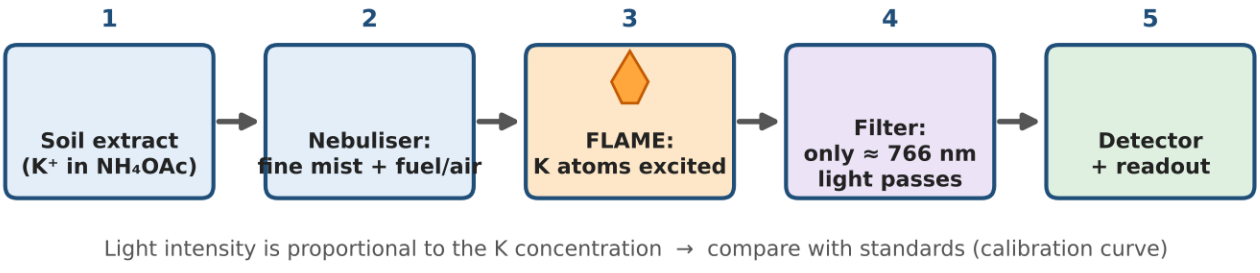



*Fig. 8.2: Inside a flame photometer (block diagram).*

## Apparatus and Reagents

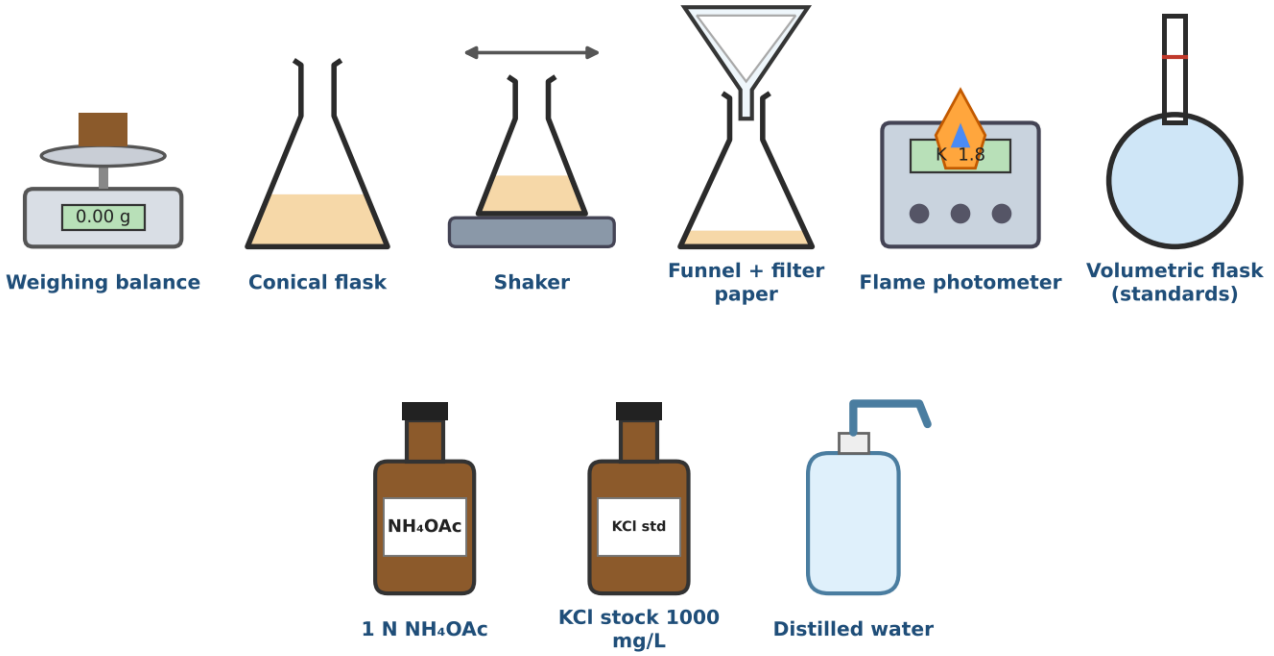



*Fig. 8.3: Apparatus and reagents. Standards: 1.907 g dried KCl per litre ≈ 1000 mg K/L stock; working standards 0, 5, 10, 15, 20 mg K/L.*

## Procedure

**A. Preparation of soil extract**

1. 5 g of air-dried soil was weighed into a clean conical flask, and 25 mL of 1 N $NH_4OAc$ solution was added.
2. The mixture was shaken for about 30 minutes, then filtered to obtain the clear soil extract.

**B. Flame photometer measurement**

1. The flame photometer was switched on and allowed to stabilise, and the K filter/wavelength (≈ 766 nm) was selected.
2. Distilled water was aspirated to zero the instrument.
3. The potassium standards (5, 10, 15, 20 mg/L) and a blank were aspirated in turn, and their readings recorded.
4. The soil extract was aspirated and its reading recorded.
5. The potassium concentration of the extract was determined from the standard calibration curve after blank correction.

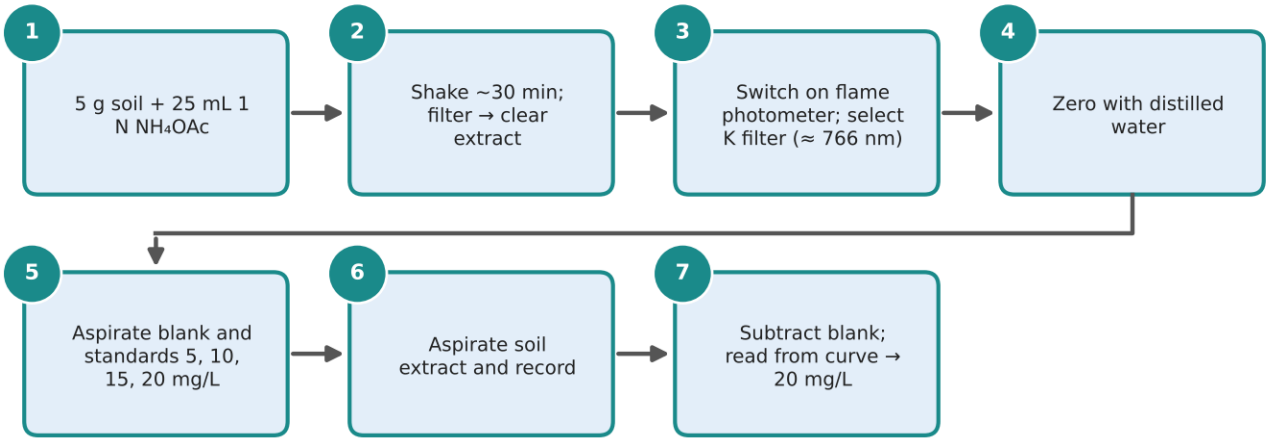



*Fig. 8.4: Procedure at a glance.*

## Observation

| No. | Solution | K concentration (mg/L) | Observed reading | Corrected reading |
|---|---|---|---|---|
| 1 | Standard 1 | 5 | 0.4 | 0.1 |
| 2 | Standard 2 | 10 | 1.0 | 0.7 |
| 3 | Standard 3 | 15 | 1.4 | 1.1 |
| 4 | Standard 4 | 20 | 2.1 | 1.8 |
| 5 | Blank | 0 | 0.3 | 0.0 |
| 6 | Soil extract | Unknown | 2.1 | 1.8 |

Corrected reading = Observed reading − Blank reading (0.3). The raw readings still contain the instrument's background signal, so only the blank-corrected values are used for quantification.

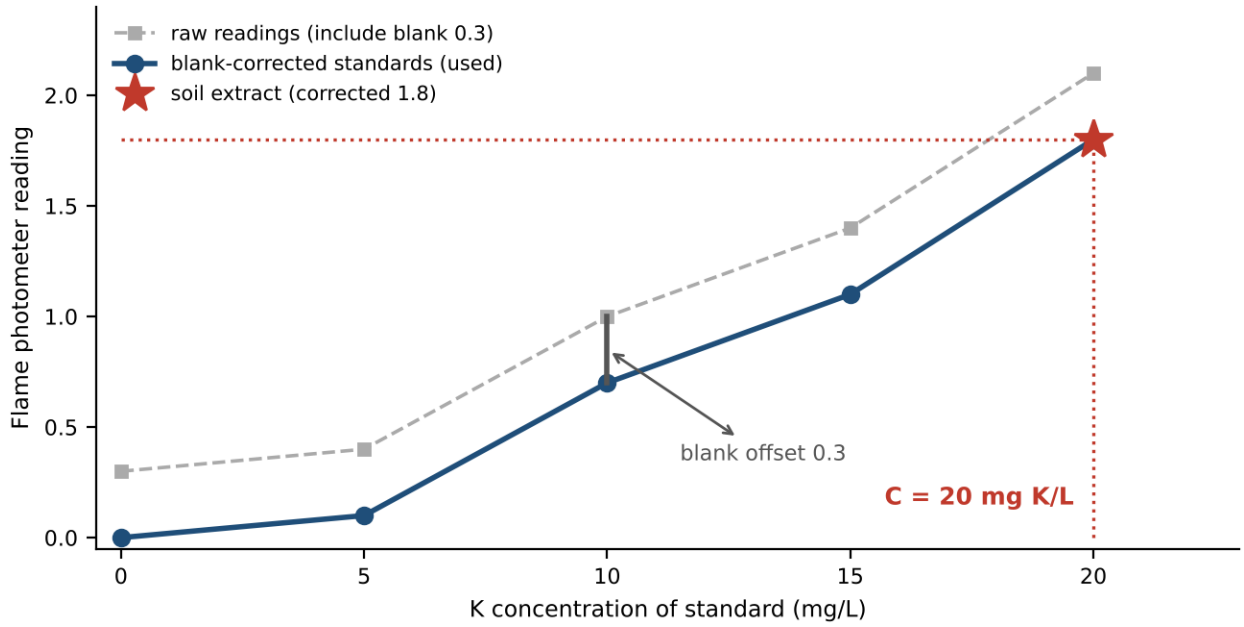



*Fig. 8.5: Blank-corrected calibration curve with the soil extract marked (raw readings shown in grey for comparison).*

## Blank Correction

*Corrected soil reading = Soil reading − Blank reading = 2.1 − 0.3 = 1.8*

The corrected soil-extract reading (1.8) coincides exactly with the corrected reading of the 20 mg/L standard (2.1 − 0.3 = 1.8), so the extract falls precisely on the calibration curve at that standard. Hence C = 20 mg K/L.

## Calculation

W = 5 g; V (extracting solution) = 25 mL; C = 20 mg/L; dilution factor D = 1.

*Available K (mg/kg) = (C × V × D) / W = (20 × 25 × 1) / 5 = 100 mg K/kg soil*

## Result

**Result:** The available potassium content of the given soil sample was found to be **100 mg K/kg soil**.

## Discussion

Using blank-corrected readings for both the standards and the extract keeps the curve and the sample on the same basis; here the corrected soil reading equals the corrected 20 mg/L standard, so no interpolation is needed. 100 mg/kg is in the low-to-medium range (below roughly 100 mg/kg is typically low, 100–200 medium, above 200 high; thresholds vary by crop and local guidelines), so potassium fertilisation should be considered for potassium-demanding crops. Error sources: incomplete extraction, glassware contamination and matrix/ionisation interference; use matrix-matched standards and replicate aspirations.

**In simple words:** Potassium glows with its own special light in a flame, and the brighter the glow the more potassium there is. We compare the glow of the soil extract with solutions of known strength to read off the amount.

# Summary: What the Results Tell Us

| TEST | RESULT | | WHAT IT MEANS |
|---|---|---|---|
| **pH** | 6.64 | 🟢 | Near neutral: close to optimum for most crops |
| **Electrical conductivity** | 125 µS/cm | 🟢 | Non-saline: no salinity stress |
| **Lime requirement** | 2.5 g $CaCO_3$/kg | 🟢 | Small: soil already near neutral |
| **Organic carbon** | 0.55 %  (OM ≈ 0.95 %) | 🟡 | On the low side: add compost / crop residues |
| **Total nitrogen** | 0.0756 % | 🔴 | Low range: needs N fertilisation / residue return |
| **Calcium** | 0.20 %  (2000 ppm) | 🟢 | Adequate base status |
| **Magnesium (Ca : Mg ≈ 2.4 : 1)** | 0.0835 %  (835 ppm) | 🟡 | Ratio below the favourable 3–6 : 1 |
| **Chloride** | 301 mg/kg | 🟢 | No toxicity expected; monitor if saline irrigation |
| **Potassium** | 100 mg/kg | 🟡 | Low-to-medium: consider K fertiliser |

🟢 good / adequate  🟡 watch / improve  🔴 low: act

*Traffic-light summary built from the interpretations given in the report's Results and Discussion sections.*

## How the experiments connect

**Soil reaction and salts (1–3):** pH 6.64, EC 125 µS/cm and a lime requirement of only 2.5 g $CaCO_3$/kg all point the same way: a near-neutral, non-saline soil that needs little correction.

**Organic matter and nitrogen (4–5):** organic carbon (0.55 %) and total N (0.0756 %) give a C : N ratio of about 7.3 : 1; both values are on the low side, so organic amendments and nitrogen management are the main needs.

**Ions and nutrients (6–8):** Ca is adequate, the Ca : Mg ratio (≈ 2.4 : 1) is a little below the favourable range, chloride is not a concern, and potassium (100 mg/kg) is at the low-to-medium boundary.

*The classification thresholds quoted here are those given in the original report; exact limits vary by crop and local guidelines.*